# Appendix - Hysteresis Test Results

Table 1. Results obtained from hysteresis for loading of 2831N without infill

| VB $[0/90/0/90]_s$ | n disc | Arrangement | Stiffness [kN/mm] | E load [J] | E unload [J] | E dissipated [J] | %E dissipated | displacement max [mm] |
|---|---|---|---|---|---|---|---|---|
| Pd2-0 | 2 | () | 1.37 | 3.03 | 2.58 | 0.45 | 14.74 | 2.79 |
| Sd4-0 | 4 | ()() | 0.74 | 5.40 | 4.76 | 0.65 | 11.95 | 4.35 |
| Pd4-0 | 4 | (()) | 2.76 | 1.72 | 1.38 | 0.33 | 19.31 | 1.67 |
| Pd6-0 | 6 | ((())) | 3.25 | 1.59 | 1.23 | 0.36 | 22.78 | 2.40 |
| Sd6-0 | 6 | ()()() | 0.51 | 8.10 | 7.17 | 0.93 | 11.53 | 6.38 |
| Hd8-0 | 8 | (())(()) | 1.50 | 3.26 | 2.71 | 0.55 | 16.91 | 3.60 |

Table 2. Results obtained from hysteresis for loading of 2681N without infill

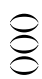
| CM $[0/90/0/90]_s$ | n disc | Arrangement | Stiffness [kN/mm] | E load [J] | E unload [J] | E dissipated [J] | %E dissipated | displacement max [mm] |
|---|---|---|---|---|---|---|---|---|
| Pd2-0 | 2 |  | 0.93 | 3.80 | 3.21 | 0.58 | 15.4 | 2.96 |
| Sd4-0 | 4 |  | 0.43 | 7.80 | 6.22 | 1.58 | 20.2 | 5.83 |
| Pd4-0 | 4 |  | 2.01 | 2.09 | 1.67 | 0.41 | 19.9 | 2.22 |
| Pd6-0 | 6 |  | 2.40 | 1.91 | 1.51 | 0.393 | 20.7 | 2.46 |
| Sd6-0 | 6 |  | 0.28 | 11.99 | 9.56 | 2.43 | 20.25 | 8.88 |
| Hd8-0 | 8 |  | 0.91 | 4.55 | 3.57 | 0.98 | 21.5 | 4.98 |

Table 3. Results obtained from hysteresis for loading of 2660N without infill

| CM [90/0/-45/45]$_s$ | n disc | Arrangement | Stiffness [kN/mm] | E load [J] | E unload [J] | E dissipated [J] | % E dissipated | displacement max [mm] |
|---|---|---|---|---|---|---|---|---|
| Pd2-0 | 2 | () | 1.22 | 3.16 | 2.74 | 0.42 | 13.15 | 1.22 |
| Sd4-0 | 4 | ()() | 0.63 | 5.98 | 4.86 | 1.12 | 18.73 | 5.37 |
| Pd4-0 | 4 | (()) | 1.72 | 2.30 | 1.86 | 0.44 | 19.21 | 2.64 |
| Pd6-0 | 6 | ((())) | 2.53 | 1.58 | 1.24 | 0.34 | 21.70 | 1.85 |
| Sd6-0 | 6 | ()()() | 0.64 | 6.63 | 5.39 | 1.24 | 18.76 | 6.13 |
| Hd8-0 | 8 | (())(()) | 1.14 | 3.61 | 2.98 | 0.63 | 17.55 | 4.08 |

Table 4. Results obtained from hysteresis for loading of 2831N with infill

| VB [0/90/0/90]$_s$ | n disc | Arrangement | Infill thickness [mm] | Stiffness [kN/mm] | E load [J] | E unload [J] | E dissipated [J] | E dissipated [%] | displacement max [mm] |
|---|---|---|---|---|---|---|---|---|---|
| Pd2-1 | | | 1 | 1.42 | 2.69 | 1.67 | 1.01 | 37.71 | 3.93 |
| Pd2-3 | 2 | () | 3 | 0.55 | 6.19 | 2.69 | 3.50 | 56.49 | 4.29 |
| Pd2-5 | | | 5 | 0.24 | 11.60 | 3.70 | 7.90 | 68.09 | 7.38 |
| Sd4-1 | | | 1 | 0.69 | 4.63 | 3.01 | 1.62 | 34.99 | 4.23 |
| Sd4-3 | 4 | ()() | 3 | 0.41 | 10.30 | 5.43 | 4.87 | 47.30 | 7.31 |
| Sd4-5 | | | 5 | 0.16 | 19.52 | 7.88 | 11.63 | 59.61 | 14.38 |
| Pd4-1 | | | 1 | 3.20 | 1.90 | 1.24 | 0.66 | 34.59 | 2.46 |
| Pd4-3 | 4 | (()) | 3 | 1.37 | 3.50 | 1.99 | 1.50 | 43.03 | 3.20 |
| Pd4-5 | | | 5 | 0.17 | 13.86 | 4.30 | 9.56 | 68.95 | 8.50 |
| Pd6-1 | | | 1 | 2.98 | 2.24 | 1.52 | 0.72 | 32.09 | 2.74 |
| Pd6-3 | 6 | ((())) | 3 | 0.78 | 5.20 | 2.67 | 2.53 | 48.71 | 4.17 |
| Pd6-5 | | | 5 | 0.23 | 12.26 | 4.54 | 7.72 | 62.94 | 8.33 |

Table 5. Results obtained from hysteresis for loading of 2681N with infill

| CM $[0/90/0/90]_s$ | n disc | Arrangement | Infill thickness [mm] | Stiffness [kN/mm] | E load [J] | E unload [J] | E dissipated [J] | E dissipated [%] | displacement max [mm] |
|---|---|---|---|---|---|---|---|---|---|
| Pd2-1 | | | 1 | 1.93 | 2.21 | 1.28 | 0.93 | 42.08 | 1.67 |
| Pd2-3 | 2 | (\|) | 3 | 0.63 | 6.04 | 2.95 | 3.09 | 51.16 | 4.12 |
| Pd2-5 | | | 5 | 0.22 | 10.97 | 3.44 | 7.53 | 68.64 | 7.50 |
| Sd4-1 | | | 1 | 0.80 | 5.16 | 3.34 | 1.82 | 35.29 | 4.40 |
| Sd4-3 | 4 | (\|)(\|) | 3 | 0.33 | 11.83 | 6.60 | 5.23 | 44.19 | 8.83 |
| Sd4-5 | | | 5 | 0.32 | 23.72 | 9.91 | 13.82 | 58.24 | 18.20 |
| Pd4-1 | | | 1 | 1.48 | 2.86 | 1.81 | 1.04 | 36.57 | 2.71 |
| Pd4-3 | 4 | ((\|)) | 3 | 1.01 | 5.22 | 2.28 | 2.94 | 56.32 | 3.82 |
| Pd4-5 | | | 5 | 0.32 | 12.86 | 3.87 | 8.99 | 69.91 | 8.03 |
| Pd6-1 | | | 1 | 2.52 | 2.12 | 1.46 | 0.66 | 31.18 | 2.29 |
| Pd6-3 | 6 | (((\|))) | 3 | 0.91 | 4.53 | 2.37 | 2.16 | 47.68 | 3.97 |
| Pd6-5 | | | 5 | 0.37 | 10.91 | 3.61 | 7.30 | 66.91 | 8.23 |

Table 6. Results obtained from hysteresis for loading of 2660N with infill

| CM [90/0/-45/45]$_s$ | n disc | Arrangement | Infill thickness [mm] | Stiffness [kN/mm] | E load [J] | E unload [J] | E dissipated [J] | E dissipated [%] | displacement max [mm] |
|---|---|---|---|---|---|---|---|---|---|
| Pd2-1 | | | 1 | 2.33 | 1.76 | 1.00 | 0.76 | 43.18 | 2.33 |
| Pd2-3 | 2 | (\|) | 3 | 0.79 | 4.72 | 2.14 | 2.58 | 54.66 | 0.79 |
| Pd2-5 | | | 5 | 0.19 | 11.56 | 3.35 | 8.21 | 71.02 | 0.19 |
| Sd4-1 | | | 1 | 0.89 | 3.98 | 2.19 | 1.79 | 44.97 | 0.89 |
| Sd4-3 | 4 | (\|)(\|) | 3 | 0.51 | 7.33 | 3.93 | 3.40 | 46.38 | 0.51 |
| Sd4-5 | | | 5 | 0.20 | 17.34 | 5.86 | 11.48 | 66.21 | 0.20 |
| Pd4-1 | | | 1 | 2.99 | 2.00 | 1.32 | 0.68 | 34.00 | 2.99 |
| Pd4-3 | 4 | ((\|)) | 3 | 0.95 | 4.20 | 1.95 | 2.25 | 53.57 | 0.95 |
| Pd4-5 | | | 5 | 0.49 | 10.03 | 3.03 | 7.00 | 69.79 | 0.49 |
| Pd6-1 | | | 1 | 2.90 | 2.04 | 1.40 | 0.64 | 31.37 | 2.90 |
| Pd6-3 | 6 | (((\|))) | 3 | 1.08 | 3.89 | 2.24 | 1.65 | 42.42 | 1.08 |
| Pd6-5 | | | 5 | 0.47 | 10.08 | 3.15 | 6.93 | 68.75 | 0.47 |